Cross-Border Competition and Tax Pass-Through: Evidence from Germany's 2022 Fuel Tax Cut

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Abstract

How does a national tax cut pass through at an international border on the treated side? How does it spill over to the untreated side? I study Germany's 2022 fuel-tax cut using every posted price on both sides of the French–German border and exact driving distances between stations. German diesel stations within 10 km of a French gas station cut prices by 0.96 cents per liter (95% confidence interval: 0.43–1.49) more than stations further than 50 km inland. That corresponds to a pass-through premium of about 6 percentage points (95% confidence interval: 2.6–8.9). The premium appears at the start of the cut, is largest within a few kilometers of the border, and disappears after the tax returns. French stations near the border also cut prices by a similar amount, but they were already falling relative to inland France before the cut.

The premium arises where the tax cut closed the cross-border price gap, that is, for diesel, and it does not appear where the gap stayed wide, that is, for gasoline. A model with treated domestic and untreated foreign rivals shows that, at a given intensity of cross-border substitution, feedback through foreign prices cannot account for the premium on its own. Instead, the evidence points to a change in that intensity: cross-border competition that switches on when a tax change closes the price gap and stays off when the gap is wide. In an integrated market, the incidence of a unilateral energy-tax change thus depends on the neighbor's prices.

Keywords: tax pass-through; tax incidence; energy taxation; cross-border competition; spatial competition; fuel prices

JEL classification: H22; H73; L13; L81; Q48; R32

1 Introduction

During the 2022 energy crisis, many European governments cut fuel taxes unilaterally to shield consumers from high prices (Drolsbach et al., 2023). How does a national tax cut reach consumers at an international border? A unilateral tax change alters both the local price level and the relative price across the frontier. In the European integrated market, consumers can buy on either side, so a border station competes with rivals whose costs remain unchanged and potentially with rivals whose costs fell. Pass-through may therefore differ between border and inland stations, and prices may move even where the tax is unchanged.

Germany's 2022 fuel discount provides the setting. From 1 June through 31 August, Germany cut its energy tax on fuel substantially. Full pass-through implied price reductions of 16.71 cents per liter for diesel and 35.16 cents for gasoline (Bundesministerium der Finanzen, 2022), when including the associated value-added-tax saving. France kept its own 18-cent tax-inclusive rebate throughout (République française, 2022a,b), so the German cut changed the cross-border price gap without a contemporaneous change in French policy. I use every posted price at German and French stations along the French–German border, along with the exact driving distances from each station to its nearest competitor across the border. The design compares German stations near a French station with German stations far inland, before and during the cut, and absorbs permanent station differences and common daily shocks via fixed effects.

The central result is a border premium that appears when the cut begins. Relative to the last week before the cut, the diesel price difference between German stations within 10 km of a French station and German stations more than 50 km away falls by 1.47 cents in the first week of the cut and by 1.98 cents in the second, and it stays negative while the cut lasts. Daily average prices place the break at the start of the cut, whether 1 June is compared with 31 May or with the preceding three weeks. Averaged over the policy period, near-border stations cut prices by 0.96 cents per liter more than inland stations (95% confidence interval 0.43–1.49), a pass-through premium of 5.8 percentage points. When the tax returned on 1 September, the premium disappeared within three weeks.

The premium is largest within a few kilometers of the border. It holds when the post-reform price change may vary with brand, motorway location, the number of nearby rivals, and urban–rural type, and with county-by-date fixed effects, which compare border stations with other stations in the same county on the same day. For the first week of the tax cut, bounds that allow post-reform trend violations as large as the largest pre-reform violation (Rambachan and Roth, 2023) exclude zero. The premium is precisely estimated with station-clustered errors and still significant at the 10% level when errors are clustered by county.

I explain the pattern across fuels and over time, presenting a simple model of cross-border competition. A foreign station constrains a border station only when consumers are close to indifferent between them, that is, when the cross-border price gap is small. Before the cut, German diesel cost about 12.5 cents more than French diesel; the cut nearly closed that gap, and the premium appeared. When the German tax returned and France raised its rebate on 1 September, German diesel again became much more expensive than French diesel, and the premium disappeared. Competition across the border was active when the gap was small and inactive when it was large. In contrast to the

diesel case, the cut turned a German disadvantage of 19–20 cents into an advantage of 11–13 cents for gasoline. The gap reversed but stayed wide, and border and inland pass-through do not differ significantly. Cross-border competition was inactive in both large gap phases.

To interpret the premium, I develop a model with a treated border station, a treated domestic rival, and an untreated foreign rival. The untreated rival lowers its price in response to the German tax cut, as French stations did with their diesel prices near Germany. This response feeds back into German pass-through only at second order. With the French response as an upper limit, the model bounds the feedback part of the premium at 0.40 of the 0.96 cents, and at most 0.78 cents. At a given intensity of cross-border substitution, feedback through foreign prices therefore cannot account for the premium on its own. However, evidence suggests a change in that intensity: cross-border competition became active when the tax cut closed the diesel price gap, and it became inactive when the gap reopened afterward. A model extension in which the intensity depends on the price gap formalizes this mechanism. A border station's markup falls when many local consumers are nearly indifferent between the two countries, producing a premium when a cut closes a wide gap and its disappearance when the gap reopens. It does not produce a premium when the gap reverses but stays wide because no consumers are nearly indifferent in either case. Calibrated to the observed gaps, the extension matches the size of the observed diesel premium and of the French response with a cross-border market of about 5 percent of local demand.

The article contributes to work on the spatial incidence of taxes. Tax differences shape where consumers buy and where retail activity locates (Manuszak and Moul, 2009; Agrawal, 2015), and cross-border fueling and shopping respond to tax-driven price differences and to distance (Rietveld et al., 2001; Banfi et al., 2005; Asplund et al., 2007; Leal et al., 2009). Pass-through of a US gasoline-tax moratorium was smaller near state borders (Doyle Jr. and Samphantharak, 2008), cigarette-tax pass-through rises with distance from a lower-tax border (Harding et al., 2012), and prices respond outside the taxing jurisdiction after a fuel-tax increase (Bajo-Buenestado and Borrella-Mas, 2019), and pass-through near regional tax borders in Spain depends on the sign of the tax differential (Iraizoz and Labeaga, 2022). For the German reform studied here, pass-through varied across regions with income and station density (Frondel et al., 2026) and over space and time (Kahl, 2024b). The article also relates to spatial competition in retail fuel markets, where the clustering of stations shapes local market power (Pennerstorfer and Weiss, 2013) and German prices do not respond to proximity to Polish stations, which Kahl (2024a) attributes to large tax and price differentials, and to the pass-through of own and rival cost shocks (Anderson et al., 2001; Weyl and Fabinger, 2013; Ritz, 2024; Muehlegger and Sweeney, 2022).

The article makes three contributions. First, it measures incidence at an international border with exceptionally detailed data: every posted price at every station on both sides of the border, which gives each station's duration-weighted daily price, and the driving distance by road from each station to its nearest rival across the border. Only quantities are missing. Studies of the same reform estimate national or regional average pass-through (Dovern et al., 2023; Kahl, 2024b; Frondel et al., 2026) or relate it to domestic market structure (Montag et al., 2023; von Waldow and Link, 2026). Within the same region, stations near a French competitor passed through 5.8 percentage points more of the cut, and the premium holds when stations are compared within the same county on the same day.

It rests on several separate contrasts: a sharp start and end of the cut, a distance gradient within a region, within-county comparisons, and a second fuel for which the model predicts no premium.

Second, the article shows that cross-border competition is not a fixed property of a border location but switches with the price gap. The same stations at the same border show a premium for diesel, where the cut nearly closed the gap, none for gasoline, where the gap reversed but stayed wide, and none for diesel once the gap reopened in September. This links two existing findings to the pricing of border stations: inactive competition at the German–Polish border, where the gap is large (Kahl, 2024a), and price-sensitive cross-border demand at this border (Adam et al., 2024). It also complements evidence that pass-through near regional tax borders depends on the sign of the tax differential (Iraizoz and Labeaga, 2022): at an international border, it depends on the size of the price gap as well.

Third, relative to work on the pass-through of own and rival cost shocks (Anderson et al., 2001; Muehlegger and Sweeney, 2022), the article develops a model in which some rivals receive the tax change and others do not, and the model signs each mechanism. The untreated rivals move their prices in the direction of the tax change, feedback through their prices raises the border station's pass-through, and replacing treated domestic rivals with untreated foreign ones lowers it. When domestic substitution is the same at the border and inland, a station's feedback premium is at most half a weighted average of the untreated rivals' responses, whatever the number and configuration of rivals. Applied to the French response, this limit shows that feedback cannot account for the premium on its own. An extension in which the intensity of cross-border substitution depends on the price gap signs the premium in each phase of the policy: positive when a cut closes a wide gap, absent or slightly negative when the gap reverses but stays wide, and gone when the gap reopens.

Section 2 describes the setting, Section 3 develops the model, and Sections 4 and 5 present the data and design. Section 6 reports the results, Section 7 assesses identification and robustness, and Section 8 concludes.

2 Institutional setting and related evidence

2.1 The German tax cut and the French rebate

The German tax cut provides an unusually clear setting with accurate data. Germany's reform temporarily lowered statutory energy-tax rates to the European minimum. Because value-added tax is levied on the tax-inclusive base, the consumer-price cost wedge fell by 16.71 cents per liter for diesel and by 35.16 cents for E5 and E10 gasoline. This is the amount equivalent to a 100% pass-through rate. The original rates returned on 1 September 2022 (Bundesministerium der Finanzen, 2022). Before April 2026, German retail prices were not administratively regulated.

France operated a separate rebate. In metropolitan France, it equaled 15 cents before tax, or 18 cents including tax, from 1 April through 31 August 2022 (République française, 2022a,b). Thus, France made no tax changes during the German tax-cut period. However, a decree issued on 22 August raised the tax-inclusive rebate to 30 cents on 1 September, the day after the last German tax cut day (République française, 2022c). TotalEnergies added its own rebate of 20 cents per liter at its

French stations from the same day (Motor1, 2022), and strikes at French refineries disrupted supply from late September (Agence France-Presse, 2022). The end of the German cut was therefore a combined event, and I present it in Section 6.3 as a reversal test that accounts for the French changes.

2.2 Cross-border choice

Let a motorist's generalized cost of buying abroad be equal to the posted foreign price plus travel time and costs. Whether the foreign station constrains a domestic seller therefore depends on both distance and the bilateral price gap, not distance alone (Kanbur and Keen, 1993; Chandra et al., 2014; Jansen and Jonker, 2018). When the gap is large, border crossing runs in one direction because the price-sensitive fraction of customers crosses the border. Then, price competition across the border can become inactive, as at the German–Polish border (Kahl, 2024a). This extensive margin is absent from the covered-market Hotelling benchmark, in which distance changes markups but need not change cost pass-through (Hotelling, 1929). First, the benchmark model below takes the intensity of cross-border substitution as given, and then Section 3.3 lets it vary with the price gap.

2.3 Evidence on the 2022 fuel discount

Published estimates generally indicate substantial average pass-through, with variation by fuel type, period, and design (Dovern et al., 2023; Drolsbach et al., 2023; Bernhardt et al., 2023; Schmerer and Hansen, 2023; Seiler and Stöckmann, 2023; Kahl, 2024b). Pass-through varied over time and space (Freitas and Syga, 2022; Kahl, 2024b), and Frondel et al. (2026) relate the regional variation to income and station density. von Waldow and Link (2026) relate pass-through to horizontal and vertical market structure, and Genakos and Pagliero (2022) show in isolated fuel markets that pass-through rises with the number of competitors. At the county level, Kahl (2024b) finds lower diesel pass-through in German counties bordering France than in counties without an external border and attributes such border-region differences to a north–south gradient. The comparison here holds the region fixed: it compares stations within the three southwestern states by driving distance to a French station. The present design asks a different question: how does incidence change when some nearby rivals receive the tax cut and others, across the border, do not?

3 A model of border tax incidence

3.1 Treated and untreated rivals

Consider a German border station G, a nearby German domestic rival D (who is sufficiently far from France), and a French rival F. For each station $j \in \{G, D, F\}$, p_j is the consumer price and q_j is the quantity sold. Let v be the value-added-tax rate, c_j the resource cost, and t_j the excise (fuel) tax; then define the tax-inclusive marginal cost as $m_j = (1+v)(c_j + t_j)$. Note that multiplying profit by $1+v$ does not change the price choice. Thus, station j maximizes $(p_j - m_j)q_j$. Let $\Delta t > 0$ denote Germany's excise-tax reduction and $\Delta\tau = (1+v)\Delta t$ its value added tax-inclusive value. The tax cut

changes both German stations' marginal costs by $-\Delta\tau$; French marginal cost remains unchanged.

Demand is linear:

$$\begin{aligned} q_G &= a_G - bp_G + b\delta_B p_D + b\rho p_F, \\ q_D &= a_D - bp_D + b\delta_B p_G, \\ q_F &= a_F - bp_F + b\rho p_G. \end{aligned} \tag{1}$$

Here a_j is a local demand intercept and $b > 0$ is the common own-price slope. The normalized slope $\delta_B \in [0, 1)$ measures substitution between the two treated German stations, while $\rho \in [0, 1)$ is what I call the intensity of cross-border substitution. The common ρ is consistent with the symmetry of an integrable quasi-linear demand system: the same marginal motorists move between G and F in either direction. Integrability with strictly concave utility requires $\delta_B^2 + \rho^2 < 1$, which also guarantees the equilibrium condition in Appendix A. The feedback calculation below implies a small ρ, but I do not estimate δ_B or δ_I (the inland counterpart of δ_B, defined in Section 3.2). The specification is a linear differentiated-products Bertrand model (Singh and Vives, 1984) applied to the incidence problem (Anderson et al., 2001), but in contrast to a two-firm model, it includes both a rival that shares the German cost shock and one that does not. Because demand curvature affects pass-through (Weyl and Fabinger, 2013), the pass-through formulas in Section 3.2 rely on linear demand and need not carry over to nonlinear demand.

The intensity ρ is a local slope, not the share of motorists who buy abroad. Let x denote driving distance to the foreign alternative and $z = p_G^e - p_F^e$ the expected German-minus-French price gap. In a smooth travel-cost model, the foreign purchase share is governed by a distribution function, whereas $\rho(x, z)$ is proportional to the density of motorists at the switching margin. It can therefore vary with both the distance and the price gap and can be hump-shaped in z. Evidence from French card transactions confirms that the switching margin responds strongly to relative fuel prices at this border (Adam et al., 2024). The benchmark holds ρ fixed; Section 3.3 lets it move with the price gap. I provide the stability condition in Appendix A.

3.2 Pass-through and the border premium

The first-order conditions, divided by b, are

$$\begin{aligned} 2p_G - \delta_B p_D - \rho p_F &= a_G/b + m_G, \\ 2p_D - \delta_B p_G &= a_D/b + m_D, \\ 2p_F - \rho p_G &= a_F/b + m_F. \end{aligned} \tag{2}$$

Let τ index the common tax-inclusive component of the two German stations' marginal costs, and let $d\tau$ be an increase in that component. When holding the local slopes fixed, solving the system gives

$$\Pi_B \equiv \frac{dp_G}{d\tau} = \frac{2 + \delta_B}{4 - \delta_B^2 - \rho^2}, \qquad \frac{dp_F}{d\tau} = \frac{\rho}{2}\Pi_B. \tag{3}$$

The first expression, Π_B, is the pass-through rate at the German border station. It measures how much the border station's price changes when the tax-inclusive marginal cost of both German stations rises by one cent. The second is the price response of the untreated French station to the same change. Prices are strategic complements, so an exposed French station lowers its price even though its own cost is unchanged (Bulow et al., 1985). The French response is first order in ρ; the feedback from F to German pass-through enters through ρ^2.

For comparison, let an inland German station face a treated domestic rival with substitution weight δ_I and no foreign rival. Its common-shock pass-through is

$$\Pi_I = \frac{1}{2 - \delta_I}. \tag{4}$$

Unlike a two-firm model with only a foreign rival, this benchmark permits pass-through between one-half and one as treated domestic competition strengthens. Comparing (3) and (4), border pass-through exceeds inland pass-through if and only if

$$\rho^2 > (2 + \delta_B)(\delta_I - \delta_B). \tag{5}$$

Equation (5) isolates two opposing forces. First, if the foreign station is additional competition and treated domestic exposure is unchanged ($\delta_B = \delta_I$), cross-border feedback raises pass-through. And second, if access to the foreign station replaces treated domestic competition ($\delta_B < \delta_I$), the border station faces fewer rivals whose costs fall, and this would lower pass-through unless cross-border feedback is dominating. Evidence that own- and rival-cost shocks pass through differently supports this distinction (Muehlegger and Sweeney, 2022).

Proposition 1 (Border incidence)**.** *With linear demand and fixed slopes, at a stable equilibrium a German tax cut lowers p_G and weakly lowers p_F, with a strict French decline when $\rho > 0$. Holding domestic competition δ_B fixed, cross-border substitution raises German pass-through because $\partial \Pi_B / \partial \rho > 0$ for $\rho > 0$. The border pass-through premium is positive if and only if* (5) *holds. If $\delta_B = \delta_I$, any $\rho > 0$ generates a positive but second-order premium; if $\rho = 0$, the premium reflects only differences in treated domestic competition and has the sign of $\delta_B - \delta_I$.*

A German border premium reflects both foreign feedback and differences in treated domestic competition and therefore does not, by itself, identify the strength of cross-border substitution. Previous tax studies find lower pass-through near a lower-tax border (Doyle Jr. and Samphantharak, 2008; Harding et al., 2012). In the model, such a pattern arises when weaker treated-rival exposure dominates foreign feedback.

How large can foreign feedback be? Suppose treated domestic competition is the same at the border and inland ($\delta_B = \delta_I = \delta$), so that any premium is due to feedback through French prices. The premium then equals the German station's response to the equilibrium French price change f, $2\rho f/(4-\delta^2)$, which already contains every round of feedback. If a station's sales fall when all prices rise together ($\delta + \rho < 1$), then the premium is at most $f/2$. The same limit holds with any number of rivals on each side and when domestic rivals have French rivals of their own (Appendix A). With the 0–10 km French estimate $f = 0.81$ cents, feedback explains at most 0.40 of the 0.96 cents; at the

upper end of the 95% confidence interval for f, at most 0.78 cents. Feedback through French prices therefore cannot account for the premium on its own.

3.3 A changing intensity of cross-border substitution

The benchmark holds ρ fixed. The reform, however, moved the price gap and with it the number of motorists close to indifferent between the two countries. To capture this margin, let $s(z)$ be the number of motorists in the border station's catchment in both France and Germany who buy in Germany when the gap is $z = p_G - p_F$, a number that decreases in the gap, $s'(z) \leq 0$. Once in Germany, these motorists choose among German stations as local motorists do, so the border station's demand is $q_G = s(z)\, L_G(p_G, p_D)$, where local demand per motorist L_G decreases in p_G. The border station chooses p_G to maximize $(p_G - m_G)\, s(z)\, L_G$. The first-order condition gives the markup

$$p_G - m_G = \frac{L_G}{-\partial L_G/\partial p_G + \varepsilon(z)\, L_G}, \qquad \varepsilon(z) \equiv -\frac{s'(z)}{s(z)} \geq 0. \tag{6}$$

The semi-elasticity $\varepsilon(z)$ of $s(z)$ is the intensity of cross-border substitution in this setting: the number of motorists at the switching margin relative to the number who buy in Germany. It is large when the gap is small, because many motorists are then close to indifferent. It is close to zero when the gap is wide in either direction, provided that many motorists in the catchment buy in Germany whatever the gap. An inland station has $\varepsilon = 0$.

Proposition 2 (A changing intensity)**.** *Hold local demand and the rivals' prices fixed. (i) The border station's markup decreases in $\varepsilon(z)$ and does not depend on the level $s(z)$. (ii) A cost cut that moves the gap from a value with $\varepsilon \approx 0$ to one with $\varepsilon > 0$ lowers the border station's price by more than an inland station's price, that is, a border premium. (iii) A cost cut that moves the gap between two values with $\varepsilon \approx 0$ leaves the border markup equal to the inland markup, whatever the change in $s(z)$, so there is no premium. (iv) When the gap widens again, the border markup rises back to the inland level, and the premium disappears.*

Part (iii) depends on how motorists who switch to Germany shop. Suppose instead that they are captive to the border station: once in Germany, they buy there whatever its price relative to other German stations. Let $C(z)$, with $C'(z) \leq 0$, denote their purchases, and normalize the number of local motorists to one, so that $q_G = L_G + C(z)$. The markup is then $(L_G + C)/(-\partial L_G/\partial p_G - C'(z))$. An inflow at a wide gap, where $C' \approx 0$, raises the markup, and the border station passes through less of the cut than inland stations. The extension therefore predicts a positive premium when a cut closes a wide gap, as for diesel, and that the premium disappears when the gap reopens, as for diesel in September. It predicts no premium, or a small negative one, when the gap reverses but stays wide, as for gasoline. The French border station faces the mirror image of $s(z)$, so it, too, lowers its markup when the gap closes. Part of the French response therefore reflects its own markup rather than the feedback, which makes the bound in Section 3.2 conservative.

Appendix A.1 derives equation (6) and computes equilibria with responding domestic and French rivals and exponentially distributed switching costs, calibrated to inland pass-through of 87 percent and to the observed pre-cut gaps. A cross-border group of 5 percent of local demand and a switching-

cost scale of 4 cents give a diesel premium of 1.07 cents, a French border response of −1.05 cents, a gasoline premium of 0.15 cents, and a September reversal of 1.13 cents, against estimates of 0.96, −0.81, about −0.65, and 0.99 cents.

3.4 Empirical predictions and scope

The model organizes the evidence into three comparisons. First, an untreated French price response in the direction of the German cost shock is the sharpest sign prediction of the benchmark. Second, at a given intensity of cross-border substitution, the German near-minus-inland coefficient measures net foreign feedback and treated domestic competition, without imposing a sign. Treated domestic competition is a feature of the station network (and not of the cross-border price gap and feedback), so a premium that reflected it alone would appear for both diesel and gasoline. Third, Proposition 2 predicts how the premium varies with the price gap across fuel types and policy phases. It predicts a positive pass-through premium when the cut closes a wide gap (June), its disappearance when the gap reopens (September), as for diesel, and no premium, or a slightly negative one, when the gap reverses but stays wide, as for gasoline. Specifically, the German gasoline price went from 19–20 cents above to 11–13 cents below the French price. Sections 6.3 and 6.4 take these predictions to the data.

Equations (3)–(5) are comparative statics at fixed slopes, and Proposition 2 adds the response that arises when the reform moves the gap and with it the intensity of cross-border substitution. My parsimonious model is used to determine the sign of the French response, bound the feedback channel, and support the qualitative predictions of Proposition 2. It is not used to fit pass-through levels or every ring coefficient. Moreover, it abstracts from several factors such as multiproduct ownership, wholesale contracts, dynamic search, and intertemporal refueling.

4 Data and price construction

4.1 Sources and sample

All German price change reports are collected by the Market Transparency Unit for Fuels, which is a unit at the German Federal Cartel Office (German *Bundeskartellamt*).[1] French price data come from the official open-data portal *prix-carburants.gouv.fr* (Ministère de l'Économie, des Finances et de la Souveraineté industrielle, 2022). The analysis covers 14 April–31 August 2022, with 48 pre-policy days and the 92-day treatment period. For the reversal test, the German panel is extended with the same construction to 26 September 2022, the day before the French refinery strikes began.

The spatial sample contains the 2,880 German gas stations in all three federal states at the French border, Baden-Württemberg, Rheinland-Pfalz, and Saarland. On the French side, it contains all 769 French gas stations in the ten départements closest to Germany. French stations more than 50 km from Germany form the control group. Each station is linked to the nearest station across the

[1] The German gasoline station and price data are made publicly available by the officially registered service provider Tankerkönig (Tankerkönig, 2022).

border by driving distance, computed with the Open Source Routing Machine (Luxen and Vetter, 2011; Giraud, 2022) on OpenStreetMap roads (OpenStreetMap contributors, 2025). Figure 1 shows why distance to an actual foreign station, rather than to the political border, is the relevant measure: stations are dense on both sides of the Rhine, but feasible cross-border alternatives are unevenly distributed along the frontier. The diesel estimation sample contains 400,250 German and 107,541 French station-days, or 507,791 observations. The E5 and E10 samples contain 437,324 and 467,452 observations.

4.2 Duration-weighted daily prices

In the observation period German stations could change prices several times a day (Haucap et al., 2017; Kahl and Wein, 2025). French gas stations change prices less often (Gautier and Le Saout, 2015). A simple mean of posted quotes would overweight intervals with frequent changes. To address this, I reconstruct each station's piecewise-constant price path from every posted price and use its duration-weighted mean from 06:00 through 22:00,

$$\bar{p}_{id} = \frac{1}{16} \int_{06:00}^{22:00} p_i(t)\, dt.$$

The primary sample requires all 16 hours to be covered. Appendix C describes the remaining construction rules in detail, and Section 7.4 reports their sensitivity.

4.3 Geography and other covariates

Distance bands are defined by driving distance to the nearest station across the border: 0–10, 10–20, 20–30, 30–40, 40–50, and more than 50 km as the inland benchmark. I also use other distance bands to assess sensitivity. Public- and school-holiday indicators capture predictable demand shifts. German stations are assigned to Eurostat's NUTS-3 urban–rural typology (Eurostat, 2026) through the municipal reference table of the Federal Institute for Research on Building, Urban Affairs and Spatial Development (Bundesinstitut für Bau-, Stadt- und Raumforschung, 2022) (Table C.3) to describe the rural-urban composition of German gas stations.

Table 1. Diesel prices before and during the German tax cut

Country	Period	Mean price	Median	SD	Holiday share	N
France	Pre	1.897	1.894	0.075	0.334	36,830
France	Post	1.990	1.985	0.110	0.609	70,711
Germany	Pre	2.022	2.014	0.073	0.201	136,990
Germany	Post	1.985	1.972	0.079	0.497	264,338

Notes: Prices are euros per liter. "Pre" is 14 April–31 May 2022; "Post" is 1 June–31 August 2022. N is station-days. The German counts include 1,078 station-days without a valid routing distance; regressions exclude them. The holiday share is the mean of an indicator for a public or school holiday.

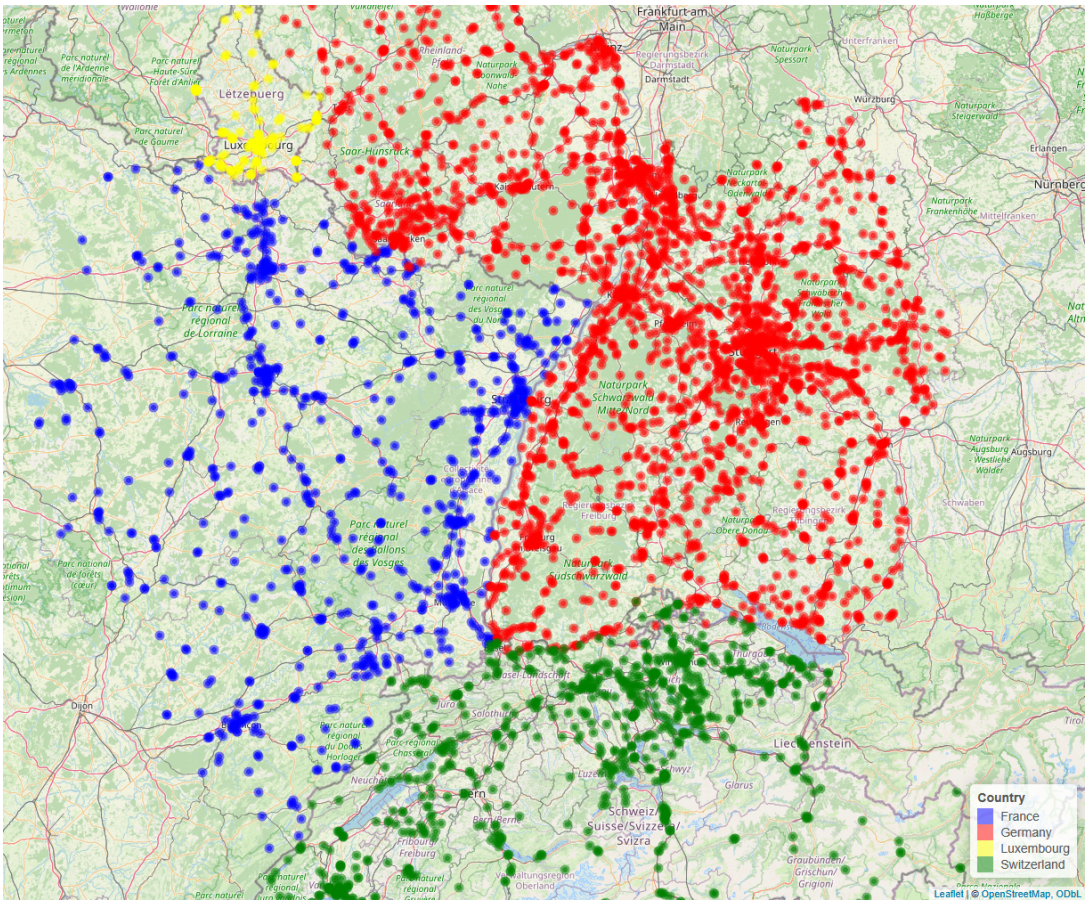


Figure 1. Fuel stations in the four-country border region

Notes: German stations are red, French stations blue, Luxembourg stations yellow, and Swiss stations green. Estimation uses German and French prices. Map lines delineate study areas and do not necessarily depict accepted national boundaries. Basemap © OpenStreetMap contributors, ODbL.

5 Empirical design

5.1 Ring specification

I follow the grouping logic of spatial difference-in-differences designs by Butts (2023). Thus, the primary regression can be formalized as

$$p_{id} = \alpha_i + \mu_d + \sum_{g \neq (FR,>50)} \beta_g (\mathbf{1}\{i \in g\} \times Post_d) + H'_{id}\chi + \varepsilon_{id}. \tag{7}$$

Here p_{id} is the duration-weighted price at station i on date d, α_i and μ_d are station and date fixed effects, and H_{id} contains only public- and school-holiday indicators. Station fixed effects absorb every time-invariant station characteristic, such as brand, motorway location, the number of nearby rivals, and urban–rural type; date fixed effects absorb shocks common to all stations, such as movements in crude-oil and wholesale prices. The eleven non-omitted groups combine country with driving-distance rings; French stations more than 50 km from Germany are the reference group. $Post_d$ equals one from 1 June through 31 August 2022. Throughout the article, standard errors are clustered by station. Additionally, I present results with clustering by county in the appendix.

For a German group, β_g is its post-policy price change relative to inland France. I report diesel

pass-through as $-\beta_g/0.167076$ and gasoline pass-through as $-\beta_g/0.351645$, hence, relative to the effective (value-added tax inclusive) tax cut. For French rings, the same normalization is only a tax-equivalent scale, and estimates in cents per liter are more intuitive. Interpreting these rates as pass-through levels requires that German and inland French prices would have moved in parallel without the cut. My main object, however, is the within-Germany contrast $\beta_{DE,0\text{–}10} - \beta_{DE,>50}$, the additional near-border response relative to inland German stations. It removes everything common to German stations, as well as any movement in the French reference group, which is a concern in contemporaneous evidence on the reform (Montag et al., 2023), and it requires parallel untreated trends only across German distance groups.

5.2 What the estimates measure

The ring coefficients in equation (7) average the near-border differential over the 92 days of the cut. Further specifications locate it in time and space, and each estimate is reported with its reference period. Event-study interactions compare the 0–10 km German ring with German stations more than 50 km from France in seven-day intervals: 1–7 June is week 0, the last pre-policy week (25–31 May) is the reference, and week 11 pools 17–31 August. I show a sensitivity check that uses the week before (18–24 May) as the reference. I also vary the distance-band size and replace 10 km with 8 or 12 km. Daily estimates are reported relative to 31 May and to the mean of the 20 preceding days, because posted prices can move before a legal date and the size of a daily break depends on the reference (Kahl and Wein, 2026). The reversal test extends the event study through the expiry of the cut, and restricted cubic splines with knots at 10, 20, 30, 40, and 50 km summarize the German distance gradient, allowing for continuous effects.

The break at the start of the cut and the reversal at its end are the most direct evidence of how border stations passed through the tax change differently. The period average summarizes the premium. Other shocks affect it only if they differ with distance to France within Germany and coincide with the policy period. Cross-border commuting, tourism, local demand, procurement, or overlapping measures that vary across the border could do so; I evaluate these threats in Section 7.

Figure 2 shows the aggregate price paths. German and French prices move broadly together before the cut. German prices then fall sharply when Germany's tax cut begins. These averages show when prices moved and by how much, but not whether stations near the border moved differently from stations farther away. My ring and event-study estimates address that question. That comparison requires parallel trends across German distance groups, not across countries. Section 7.1 rejects strict parallel pre-trends, although the leads do not drift toward the post-reform level. Bounds that allow for violations of parallel trends (Rambachan and Roth, 2023) support the break at the start of the cut but, unsurprisingly, not the average over the policy period.

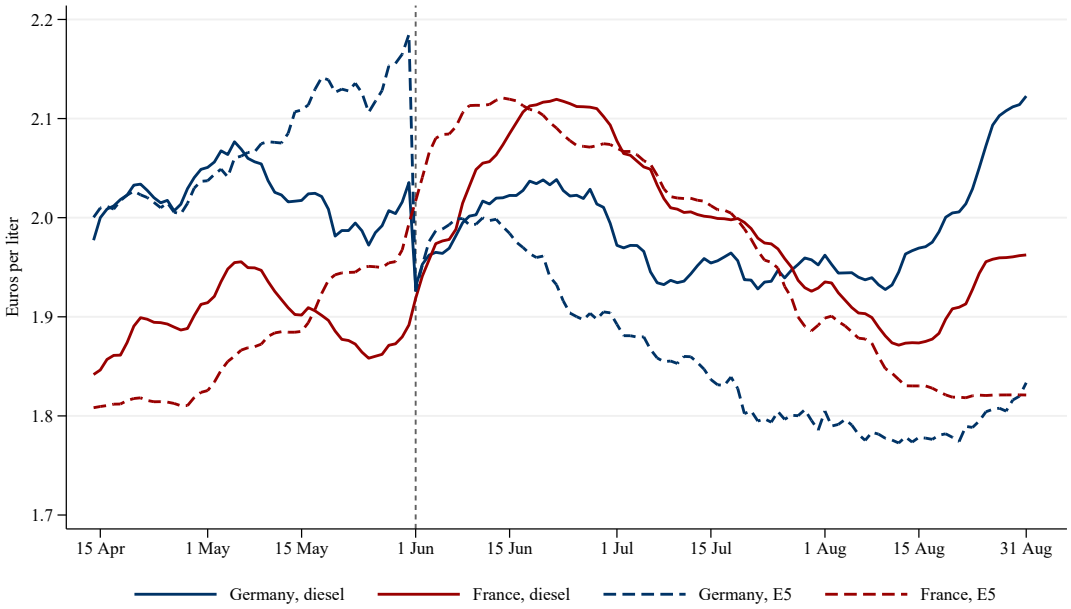


Figure 2. Average diesel and E5 prices of the French and German sample stations

Notes: Daily averages across the sample stations in each country (three German states and ten French départements), in euros per liter. The vertical line marks 1 June 2022, the first day of the German tax cut.

5.3 Overlapping measures

Other measures overlapped with the reform, and most are absorbed by the design. Measures common to all German stations drop out of the within-Germany contrast: the 9-euro public-transport ticket of June to August (Krämer et al., 2022) and the levying of the energy tax when fuel leaves the tax warehouse, so that stations entered June with stocks taxed at the old rate (Bundesregierung, 2022). France-wide measures, such as extending the French rebate from 31 July to 31 August (République française, 2022b), move the French reference group, which the within-Germany contrast does not use; their effect on German border stations runs through the cross-border price gap the analysis studies. Regional shocks, such as low water on the Rhine in August (Bundeskartellamt, 2022), are absorbed by county-by-date fixed effects in a robustness check, and a further check ends the policy period on 16 August. Two measures could differ with distance to France. First, Luxembourg reduced fuel prices by 7.5 cents per liter from 13 April through 31 August (Gouvernement du Grand-Duché de Luxembourg, 2022; L'essentiel, 2022), which could affect German stations near both borders. I address this by dropping German stations near other borders. Second, Germany notified flexible controls at its internal borders, including the French border, from 13 June to 3 July (European Commission, 2026), which began after the premium appeared and ended while it persisted. Neither measure changes around 1 June, so neither can produce the break at the start of the cut.

6 Results

6.1 The break when the cut begins

Figure 3 compares German diesel stations within 10 km of a French station with German stations more than 50 km inland. Relative to the last pre-policy week, the differential turns negative when the cut begins: it equals -1.47 cents in week 0 and -1.98 cents in week 1 and remains negative throughout the policy period. Ten of twelve post-reform coefficients are individually significant at 5%, and they are jointly significant ($p < 0.001$). The break does not depend on the reference week: with 18–24 May as the reference, the final pre-policy week is 0.24 cents ($p = 0.366$; Fig. B.6). Reassuringly, the same profile appears with 8 km and 12 km cutoffs (Fig. B.5).

Daily prices place the break at the start of the tax cut. On 1 June the near-border differential is -1.35 cents (SE 0.47) relative to 31 May and -1.95 cents (SE 0.37) relative to the mean of the 20 preceding days (Fig. B.7). The two numbers differ because the differential on 31 May was itself low. Specifically, most pre-reform days lie 0.5–1.2 cents above it. As for any estimate dated by a statute, the size of the daily break depends on the reference (Kahl and Wein, 2026).

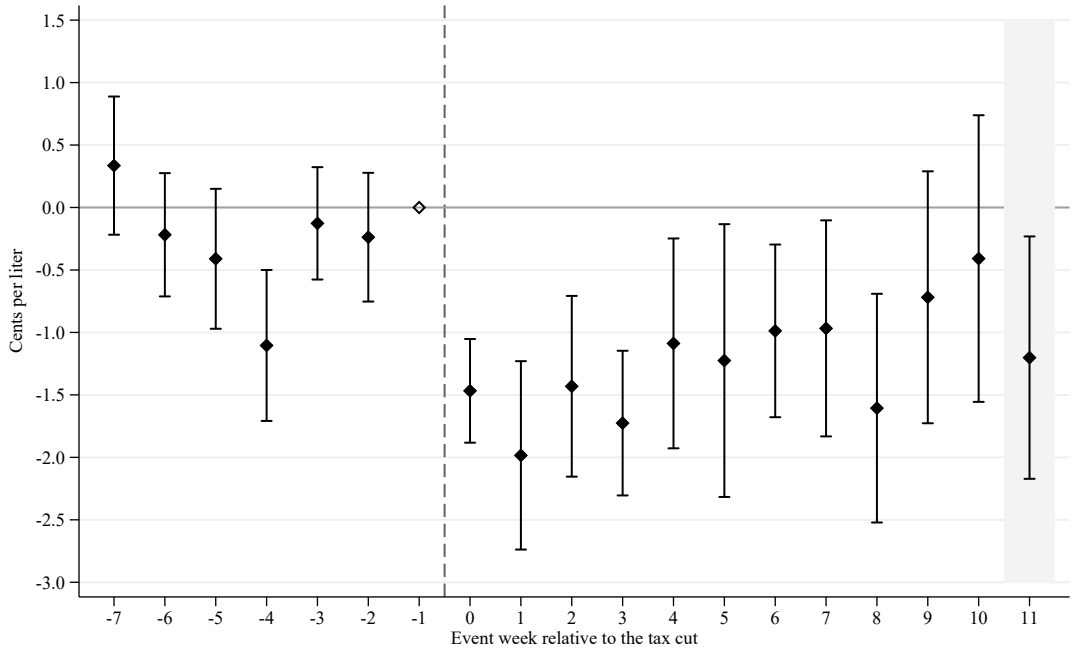


Figure 3. Event study: German 0–10 km diesel ring relative to inland Germany

Notes: Coefficients in cents per liter compare German stations within 10 km of a French station with German stations more than 50 km away. Event week 0 is 1–7 June 2022; the shaded week 11 pools 17–31 August. The omitted week −1 (25–31 May) is shown as a hollow marker. Bars are 95% confidence intervals based on station-clustered standard errors. Five of six leads are individually insignificant; the six leads are jointly significant ($F = 11.05$, $p < 0.001$).

6.2 The premium over the policy period

Table 2 averages the response over the policy period. Relative to inland France, German diesel prices fall by 14.18 cents per liter within 10 km of a French station, 84.9% of the tax-inclusive cut, and by 13.22 cents, or 79.1%, more than 50 km away. Thus, the near-border premium is 0.96 cents per liter (95% CI: 0.43 to 1.49; $p < 0.001$), or 5.8 percentage points of the cut (95% CI: 2.6 to

8.9). Because the differential is negative in every policy week (Figure 3), this average summarizes a premium that lasts through the cut rather than a mixture of episodes. Weekly estimates against French stations vary with the national adjustment path, but the German rings move together (Fig. B.3).

The premium is largest close to the border. In the German-only sample, where it is -0.97 cents, near-border cutoffs of 4, 6, 8, 10, 12, 15, and 20 km give -1.70, -0.98, -0.93, -0.97, -1.07, -0.67, and -0.37 cents. Moreover, driving-time cutoffs give the same profile (Table C.6). Rings farther out show no clear pattern: 20–30 and 30–40 km rings exceed the inland response, the 10–20 km ring is close to it, and the 40–50 km ring lies in between (Figure 4).

French prices near the border also fall: relative to French stations more than 50 km from Germany, diesel prices fall by 0.81 cents within 10 km and by 0.73 to 1.14 cents in the 10–40 km rings. That is exactly the sign the model predicts for an untreated rival. Because French border prices were already falling relative to inland France in May (Section 7.1), these estimates serve only as an upper bound on the German-caused response.

With that upper bound, the model of Section 3 limits feedback through French prices, at a given intensity of cross-border substitution, to at most 0.40 of the 0.96 cents. The premium therefore reflects how border and inland markets absorbed the cut; as Proposition 2 predicts, it arises where the cut closed the cross-border price gap (Section 6.4). Finally, the treated domestic competition and the number of motorists close to indifferent both vary across local markets, which fits the nonmonotone ring profile.

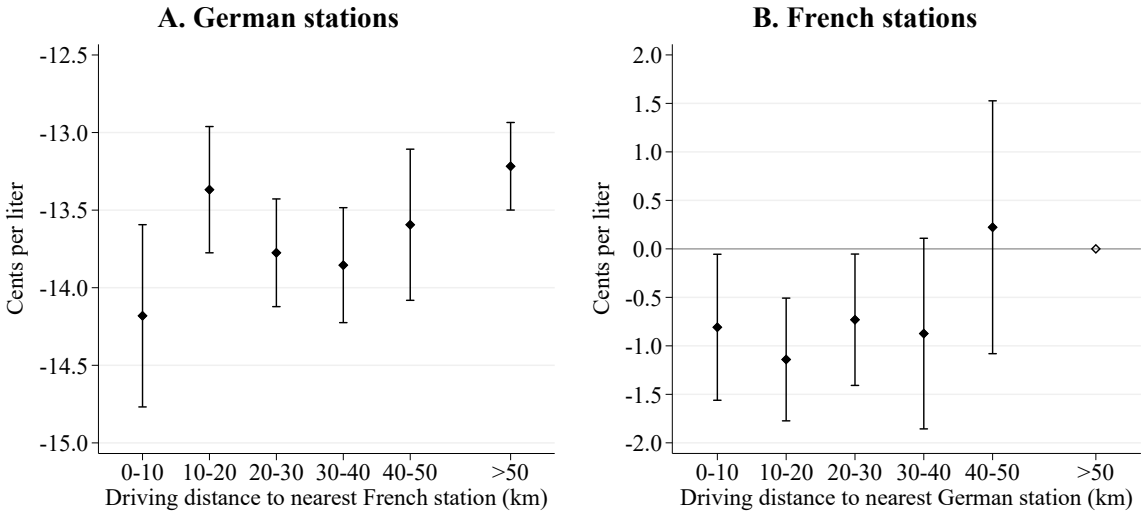


Figure 4. Diesel price responses by distance on both sides of the border

Notes: Points are the Table 2 coefficients in cents per liter; bars are 95% confidence intervals based on station-clustered standard errors. Both panels are relative to French stations more than 50 km from Germany, the omitted group shown as the hollow marker in Panel B. Vertical scales differ across panels.

Table 2. Spatial price responses to the German fuel-tax cut: diesel

Country and distance	Price change (cents/liter)	SE	*p*	Tax-equivalent response (%)
Germany, >50 km	-13.22	0.14	<0.001	79
Germany, 0–10 km	-14.18	0.30	<0.001	85
Germany, 10–20 km	-13.37	0.21	<0.001	80
Germany, 20–30 km	-13.77	0.18	<0.001	82
Germany, 30–40 km	-13.85	0.19	<0.001	83
Germany, 40–50 km	-13.59	0.25	<0.001	81
France, 0–10 km	-0.81	0.38	0.035	5
France, 10–20 km	-1.14	0.32	<0.001	7
France, 20–30 km	-0.73	0.35	0.035	4
France, 30–40 km	-0.87	0.50	0.082	5
France, 40–50 km	0.22	0.67	0.738	-1

Notes: The dependent variable is reported in cents per liter. The omitted group is French stations more than 50 km from Germany. The model includes station and date fixed effects plus public- and school-holiday controls. Standard errors are clustered by station. The final column divides the negative coefficient by the 16.7076-cent tax-inclusive diesel cut; for French rings it is a tax-equivalent response, not French tax pass-through. N = 507,791.

6.3 When the tax returns

If the premium reflects how border stations pass through German tax changes, it should disappear when the tax cut is reverted on 1 September. The French changes on the same day (Section 2.1) work against such a reversal. If anything, lower French prices give German border stations a reason to raise prices by less than inland stations.

Figure 5 extends the event study to 26 September, before the French refinery strikes. The weekly estimates are imprecise and none is individually significant: the point estimates are -1.05 cents in the first week of September, -0.51 cents in the second, and +0.29 and +0.41 cents in the third and fourth. Over 1–26 September the premium averages 0.01 cents, against -0.98 cents during the cut; the change of +0.99 cents has a station-clustered *p*-value of 0.053. The premium is therefore tied to the policy period, and it disappears once German diesel became more expensive than French diesel, as Proposition 2 predicts.

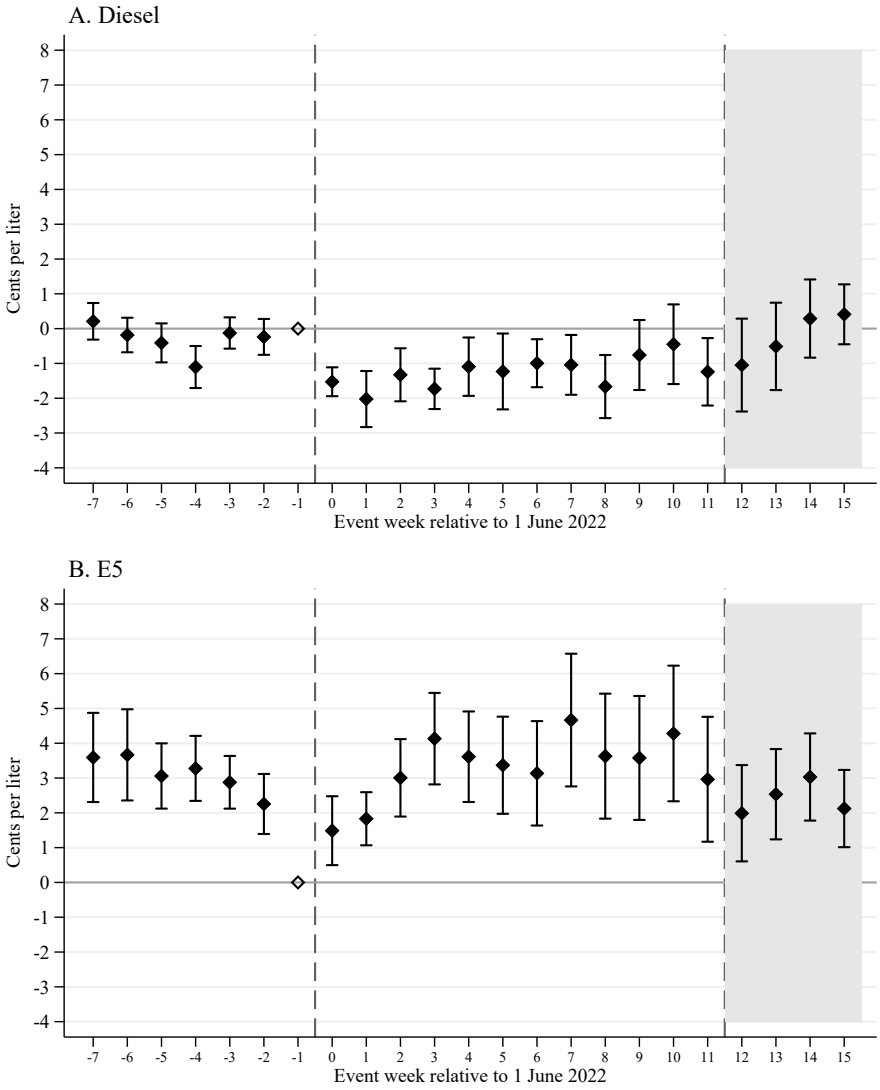


Figure 5. Event study through the expiry of the cut: German stations within 10 km of a French station relative to inland Germany

Notes: Coefficients in cents per liter for diesel (Panel A) and E5 (Panel B). Event week 0 is 1–7 June 2022; week 11 pools 17–31 August; weeks 12–15 (shaded) cover 1–26 September, after the German tax returned. The omitted week −1 (25–31 May) is shown as a hollow marker; dashed lines mark 1 June and 1 September. Station and date fixed effects and holiday controls; bars are 95% confidence intervals based on station-clustered standard errors. N = 380,665 station-days and 2,309 stations (diesel); N = 377,013 station-days and 2,287 stations (E5).

6.4 Gasoline

German gasoline pass-through is high across the distance distribution: for E5 it is 89.7% in the border ring and 91.5% inland, and for E10 88.9% and 90.8% in the rings, respectively. The nearest-ring responses are 1.8 and 1.9 percentage points lower, and neither difference is statistically significant ($p = 0.197$ for E5; $p = 0.176$ for E10). The diesel premium therefore does not arise mechanically from proximity.

The fuels differ in how the cut moved the cross-border price gap. Germany was initially more expensive than France by about 12.5 cents for diesel and 19.8 and 19.0 cents for E5 and E10, respec-

tively. During the cut, the diesel gap nearly closed, whereas the gasoline gaps reversed to roughly -11 to -13 cents (Figs. B.1 and B.2). For diesel, German and French stations became close substitutes for motorists near the border, and cross-border competition became active. For gasoline, the gap reversed but stayed wide. Motorists who had crossed to France now bought at home, and French motorists came to Germany, but few were close to indifferent in either regime.

As Section 3.3 shows, such a change in the customer base lowers border pass-through only if the arriving motorists are captive to border stations, that is, if they buy there without comparing prices at other German stations. Border stations then keep part of the cut as a higher markup and lower their prices by less than inland stations. The gasoline estimates point in that direction but are small: near-border stations cut E5 prices by 0.57 cents less than inland stations in June and, when the tax returned, raised them by 1.07 cents less in September. Dropping motorway stations or stations near Luxembourg, Switzerland, and Belgium leaves the signs of both E5 responses unchanged.

Within the policy period, week-to-week fluctuations in the diesel gap do not predict the size of the premium. The evidence for this account comes from the contrast across fuels and from the end of the cut, not from weekly variation. Within 40 km of Germany, French gasoline prices also fall relative to inland France, the sign the benchmark predicts for an untreated rival. The coefficients are unevenly ordered by distance, and, as for diesel, I refrain from interpreting their magnitudes (Tables C.4 and C.5). Restricted cubic splines support the border–inland contrast: diesel pass-through is highest at the border and gasoline pass-through is lowest there. In between, the curves show no clear pattern (Fig. B.4).

7 Identification and robustness

7.1 Pre-trends and bounds

Five out of the six weekly leads in Figure 3 are individually insignificant; the exception is week -4, at -1.10 cents. The leads are jointly significant ($F = 11.05$, $p < 0.001$), also when week -4 is left out ($p < 0.001$), so I reject strict parallel pre-trends. However, the leads show no clear trend (a linear pre-trend is -0.01 cents per week, SE 0.06), so the premium does not simply continue a difference that was already growing before the cut. Bounds that allow post-reform violations of parallel trends (Rambachan and Roth, 2023; Roth, 2022) make the distinction precise. If I allow post-reform violations to be as large as the largest pre-reform violation (relative magnitude $\bar{M} = 1$), the 95% interval for the first policy week is -2.89 to -0.02 cents. The break at the start of the cut is therefore robust to such violations. The average over the policy period is not robust to this type of inference sensitivity check. Its robust interval includes zero at $\bar{M} = 0.25$, the smallest violation that I considered. This partly reflects how the bounds are built. The allowed violations accumulate week by week, so the interval widens with the horizon, and a single pre-reform move sets its size. At $\bar{M} = 1$, the near-minus-inland difference could drift by about one cent in every policy week, more than 10 cents by mid-August, although it stayed within about one cent of zero before the tax cut. A steady drift of this kind would also continue after the tax returned, whereas the premium averages zero in September (Section 6.3). The average is, in addition, more exposed to overlapping shocks

than the first week (Section 5.3).

Assigning a false treatment date of 18 May yields +0.61 cents for the nearest German ring (p = 0.050), opposite in sign to the premium. The same placebo gives negative coefficients of 1.0 to 1.9 cents for four French rings (Table C.7). As mentioned above, French border prices were already falling relative to inland France before the reform, which is why I abstain from interpreting the French estimate magnitudes. However, despite the pre-trend, the negative sign of the French border area estimate is in line with what the model predicts.

7.2 Composition and local shocks

Table C.6 reports the premium under alternative specifications in the German-only sample, where the baseline estimate is -0.97 cents. When the post-reform change may vary with the pre-reform price, the five largest brands, motorway location, urban–rural type, and the number of German rivals within 5 km, the premium is -0.76 cents. County-by-date fixed effects, which absorb every shock common to a county, give -0.79 cents, and fixed effects for government districts (*Regierungsbezirke*) by date give -1.31 cents. Dropping motorway stations gives -0.95 cents, dropping stations within 20 km of Luxembourg, Switzerland, or Belgium gives -0.83 cents, and ending the policy period on 16 August yields an estimate of -0.97 cents. Against stations 10–50 km from France instead of inland stations, the premium is -0.54 cents (p = 0.050). A triple difference that compares diesel with E5 at the same stations, and so absorbs every station-by-date shock common to both fuels, gives -1.58 cents (station-clustered SE 0.51), or -7.5 percentage points of pass-through. Overall, the premium keeps its sign across these specifications, from -0.54 to -1.58 cents.

7.3 Inference

Distance to France is assigned station by station, so the main estimates cluster by station. Fuel prices can also move together within counties, and Table C.8 reports alternatives for the German-only premium of -0.97 cents. Clustering by the 86 counties roughly doubles the standard error, to 0.54 cents (p = 0.078), and a county wild cluster bootstrap gives p = 0.088. Spatial-HAC standard errors (Conley, 1999) with cutoffs between 10 and 100 km give p-values between 0.05 and 0.18. The premium is therefore significant at conventional levels with station clustering and at the 10% level with county clustering. The September change in Section 6.3 is less precise with county clustering (p = 0.38).

7.4 Price construction

Results are similar when the daily price averages the full day, which gives 83.5% pass-through in the nearest German ring and 78.0% inland. Moreover, estimates are comparable when carried-forward prices are capped at 3, 7, or 14 days, and when ambiguous French timestamps are resolved by the first or the last record (Tables C.9 and C.10).

7.5 Scope

The data measure posted prices, not quantities, margins, station-level wholesale costs, or consumer trips. The design therefore cannot separate cross-border shopping from correlated local demand, procurement, or market-structure changes. Driving distance is measured precisely, but the nearest foreign station need not be every motorist's relevant alternative. The active-competition account rests on the contrast across fuels and on estimates from the beginning and the end of the tax cut. Future research may look at quantity or travel data to measure cross-border switching directly. The article identifies spatial price responses and assesses whether they fit competition across the border. It does not estimate welfare or a structural demand system.

8 Conclusion

National average incidence estimates hide a local response to Germany's 2022 fuel-tax cut at the border. When the cut began, the diesel price difference between German stations within 10 km of a French station and inland German stations fell by 1.47 cents in the first week and by 1.98 cents in the second. Over the policy period, near-border stations cut prices by 0.96 cents per liter more than stations more than 50 km inland. That corresponds to 5.8 percentage points of the tax-inclusive cut. The differential is largest within a few kilometers of the border, holds with county-by-date fixed effects and when the post-reform change may vary with station characteristics, and disappears after the tax returns. Gasoline, for which the cross-border gap stayed wide, shows no such premium. The jointly rejected pre-trend test and the wider confidence intervals under county clustering counsel against treating 0.96 cents as a precise structural parameter, but the timing, the distance gradient, and the reversal tie the differential to the tax cut.

A model with a treated border station, a treated domestic rival, and an untreated foreign rival shows that, at a given intensity of cross-border substitution, feedback through foreign prices cannot account for the premium on its own. The pattern across fuels and over time matches an extension in which that intensity depends on the price gap. Where the tax cut brought German and French prices close together, as for diesel, competition across the border became active, and border stations passed through more of the cut. Where the gap stayed wide, as for gasoline, or reopened, as for diesel in September, competition was inactive, and the premium was absent.

For tax policy, a national fuel-tax change reaches consumers unevenly within the country, and the variation depends on the fuel type and on how the change moves the cross-border price gap. Evaluations that average over all gas stations miss it. Because the same tax change can switch cross-border competition on for one product and leave it off for another, the incidence of a unilateral tax change at a border depends on the neighbor's prices as well as on the domestic market. This matters for the many unilateral fuel-tax cuts of 2022 (Drolsbach et al., 2023) and for energy-tax coordination in the European Union: where national fuel taxes differ, the incidence of a national tax change depends on the neighbor's tax level.

CRediT authorship contribution statement

Mats Petter Kahl: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of competing interest

Mats Kahl has, since October 2026, provided paid consulting services to Compass Lexecon, an economic consultancy, on competition matters, through a research-transfer agreement with Universidad Carlos III de Madrid. The research in this article was conceived and carried out independently of that engagement; Compass Lexecon and its clients had no role in the study design, data collection, analysis, interpretation, writing, or the decision to submit, and the article uses no confidential information obtained through the engagement.

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Data availability

The German station-level fuel price data were provided by Tankerkönig (www.tankerkoenig.de), based on data from the Market Transparency Unit for Fuels (Markttransparenzstelle für Kraftstoffe, MTS-K) of the Bundeskartellamt, under a CC BY-NC-SA 4.0 licence; they can be obtained free of charge after registration. The French price data are open data (Licence Ouverte 2.0). All code, documentation, and other inputs, including a script that downloads the raw files and a script that rebuilds every table, figure, and in-text number from them, are available on Zenodo at https://doi.org/10.5281/zenodo.23425723 (Kahl, 2026). The derived station-level panel is available from the author on reasonable request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used artificial intelligence tools (Claude (Anthropic), Codex (OpenAI), Grammarly, and Cursor (Anysphere)) to draft and edit the manuscript text, to write, run, and check the analysis code under the author's specifications, and to check numbers and citations against their sources. After using these tools, the author reviewed, verified, and edited all text, code, and results as needed, and takes full responsibility for the content and integrity of the

published article.

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A Model derivations

Dividing the firms' first-order conditions by b yields equation (2). Strict own-price concavity follows immediately because

$$\frac{\partial^2 \pi_j}{\partial p_j^2} = -2b < 0 \qquad (j \in \{G, D, F\}).$$

The coefficient matrix of the price system is

$$A = \begin{pmatrix} 2 & -\delta_B & -\rho \\ -\delta_B & 2 & 0 \\ -\rho & 0 & 2 \end{pmatrix}, \qquad \det(A) = 2(4 - \delta_B^2 - \rho^2).$$

Thus a unique interior equilibrium exists under the maintained stability condition

$$\Delta_B \equiv 4 - \delta_B^2 - \rho^2 > 0, \tag{A.1}$$

together with positive equilibrium quantities. A sufficient primitive restriction is $\delta_B, \rho \in [0, 1)$, which makes the best-response map a contraction. Integrable demand with strictly concave quasi-linear utility requires the stronger condition $\delta_B^2 + \rho^2 < 1$, under which the matrix of demand slopes in equation (1) is negative definite; this condition implies $\Delta_B > 0$. Nonsingularity alone requires $\Delta_B \neq 0$; the stronger condition (A.1) fixes the economically relevant signs of the comparative statics. Symmetry of the off-diagonal G–F terms follows from the integrable local demand system in equation (1). In the two-firm special case, restoring the unnormalized cross-price slope $\gamma = b\rho$ gives the first-order-condition matrix

$$\begin{pmatrix} 2b & -\gamma \\ -\gamma & 2b \end{pmatrix},$$

whose determinant is $4b^2 - \gamma^2$. The expression is therefore the determinant condition for a unique local price equilibrium, not an additional behavioral assumption.

To derive pass-through, let $g = dp_G/d\tau$, $d = dp_D/d\tau$, and $f = dp_F/d\tau$, where $d\tau$ raises both German marginal costs by one unit. Differentiating the first-order conditions gives

$$\begin{aligned} 2g - \delta_B d - \rho f &= 1, \\ 2d - \delta_B g &= 1, \\ 2f - \rho g &= 0. \end{aligned} \tag{A.2}$$

The second and third equations imply $d = (1 + \delta_B g)/2$ and $f = \rho g/2$. Substitution into the first gives

$$g = \frac{2 + \delta_B}{4 - \delta_B^2 - \rho^2}, \qquad f = \frac{\rho}{2} g,$$

which proves equation (3). For an inland German station and its treated rival, the derivative system is $2i - \delta_I j = 1$ and $2j - \delta_I i = 1$. Symmetry gives $i = j = 1/(2 - \delta_I)$, proving (4).

Finally, $\Pi_B > \Pi_I$ is equivalent, under positive denominators, to

$$(2 + \delta_B)(2 - \delta_I) > 4 - \delta_B^2 - \rho^2,$$
$$\rho^2 > (2 + \delta_B)(\delta_I - \delta_B),$$

which is condition (5). The foreign response $f = (\rho/2)g$ is linear in ρ around $\rho = 0$, whereas

$$\frac{\partial \Pi_B}{\partial \rho} = \frac{2\rho(2 + \delta_B)}{(4 - \delta_B^2 - \rho^2)^2}$$

equals zero at $\rho = 0$. Hence foreign feedback affects German pass-through only at second order near zero. If $\delta_B = \delta_I$, condition (5) reduces to $\rho^2 > 0$; if $\rho = 0$, $\Pi_B = 1/(2 - \delta_B)$, so the premium has the sign of $\delta_B - \delta_I$.

Finally, equation (3) differentiates an equilibrium at fixed slopes. If ρ varies with the price gap z, the total derivative is $dp_G/d\tau = \Pi_B + (\partial p_G/\partial \rho)\, \rho'(z)\, dz/d\tau$, and the linear benchmark does not sign the second term. Appendix A.1 derives it for the demand system of Section 3.3.

The feedback limit. With $\delta_B = \delta_I = \delta$, equations (3) and (4) give a premium of $\rho^2 \Pi_B/(4 - \delta^2)$. Because $f = \rho \Pi_B/2$, this equals $2\rho f/(4-\delta^2)$, and $\rho < 1-\delta$ bounds it by $2(1-\delta)f/(4-\delta^2) \le f/2$. The limit extends to any network of stations. Let S collect the slopes between German stations, with equal row sums δ, let $R \ge 0$ collect the slopes towards French stations, with row sums of at most $1-\delta$, and let $f \ge 0$ be the vector of French responses. The first-order conditions give $(2I - S)g = \mathbf{1} + Rf$, and an inland station has $g = 1/(2 - \delta)$. The premium vector $e = g - \mathbf{1}/(2 - \delta)$ therefore solves $(2I - S)e = Rf$. Since $(2I - S)^{-1} \ge 0$ with row sums $1/(2 - \delta)$, each e_i is a weighted sum of the French responses with weights that sum to at most $(1 - \delta)/(2 - \delta) \le 1/2$. A station's premium is thus at most half a weighted average of the French responses.

A.1 A changing intensity of cross-border substitution

Let $q_G = s(z)\, L_G(p_G, p_D)$ with $z = p_G - p_F$. The first-order condition for p_G is

$$s\, L_G + (p_G - m_G)(s' L_G + s\, \partial L_G/\partial p_G) = 0.$$

Dividing by s and writing $\varepsilon = -s'/s$ gives equation (6). With local demand and the rivals' prices held fixed, the markup decreases in ε and does not depend on s. A change in costs that raises ε therefore lowers the border price relative to an inland station with $\varepsilon = 0$, a change that leaves ε near zero leaves the two markups alike, and a change that lowers ε again reverses the first; these are parts (i)–(iv) of Proposition 2. With captive arrivals, $q_G = L_G + C(z)$, the first-order condition is $L_G + C + (p_G - m_G)(\partial L_G/\partial p_G + C') = 0$, which gives the markup in Section 3.3.

To check the proposition with responding rivals, I compute Nash equilibria for a pair of German

border stations, a pair of French border stations, and German and French inland pairs. Each pair competes locally with linear demand $L_i = A - p_i + \delta p_j$. With $\delta = 0.85$, inland pass-through is $1/(2-\delta) = 87$ percent; A is set so that the inland markup is 12 cents. Masses ν of German and of French border motorists, relative to local demand, switch country when the gap exceeds their switching cost, which is exponentially distributed with mean λ cents. The number of motorists buying in Germany is $s(z) = 1 + \nu e^{-z/\lambda}$ for $z \geq 0$ and $s(z) = 1 + 2\nu - \nu e^{z/\lambda}$ for $z < 0$, and France mirrors it. French costs are set so that the pre-cut gaps are 12.5 cents for diesel and 19.8 cents for gasoline. The cut lowers German costs by 16.71 and 35.16 cents; in September the German tax returns and the French rebate rises by 12 cents. Table A.1 reports selected parameter values. Across 36 combinations, with $\delta \in \{0.80, 0.85\}$, $\nu \in \{0.02, 0.05, 0.10\}$, $\lambda \in \{2, 4, 8\}$, and both demand versions, the diesel premium is positive (0.01 to 2.68 cents) and reverses in September by about the same amount. The gasoline premium lies between 0.01 and 0.58 cents when arriving motorists shop like local ones and between −0.76 and −0.09 cents when they are captive. The sizes depend on the functional form; they illustrate the mechanism and are not estimates.

Table A.1. Equilibrium illustration of a changing intensity of cross-border substitution

Arriving motorists	ν	λ	Diesel premium	French response	Gasoline premium	September reversal
Shop like locals	0.02	2	0.59	-0.57	0.01	0.59
Shop like locals	0.05	4	1.07	-1.05	0.15	1.13
Shop like locals	0.10	8	1.06	-1.00	0.44	1.29
Captive	0.05	4	0.17	-0.61	-0.29	0.18
Captive	0.10	2	0.87	-1.83	-0.68	0.88
Estimates			0.96	-0.81	-0.63 (E5), -0.67 (E10)	0.99

Notes: Cents per liter; $\delta = 0.85$. The premium is the additional price cut at German border stations relative to German inland stations. The French response is the price change at French border stations relative to French inland stations. The September reversal is the additional price increase at German border stations when the tax returns and the French rebate rises. The estimates are from Table 2, Tables C.4 and C.5, and Section 6.3; the French estimate is an upper bound (Section 7.1). Computed by `code/model_illustration.py` in the replication package.

B Additional figures

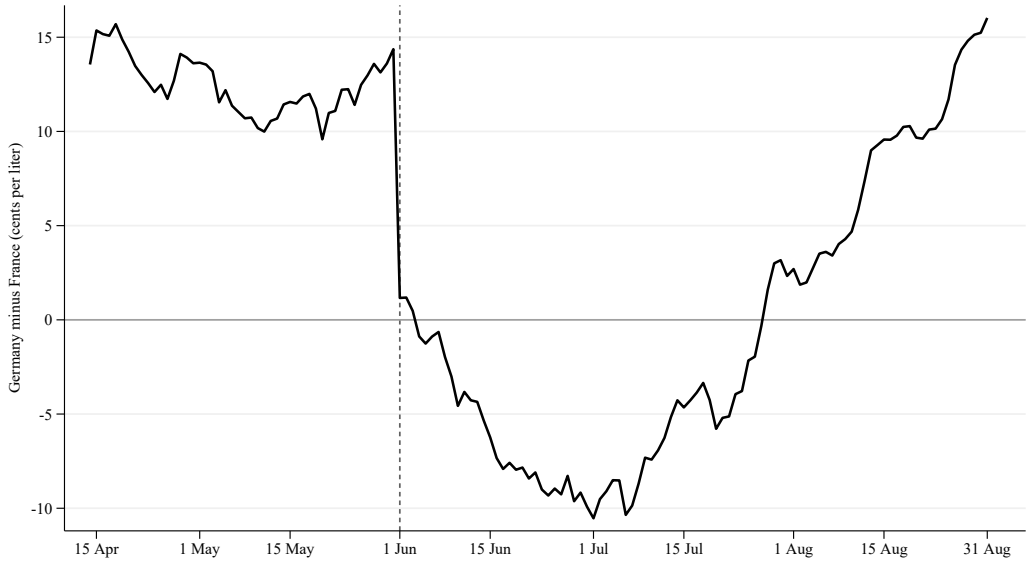


Figure B.1. Diesel-price difference between Germany and France

Notes: Daily average price of the German sample stations minus that of the French sample stations, in cents per liter. The vertical line marks 1 June 2022.

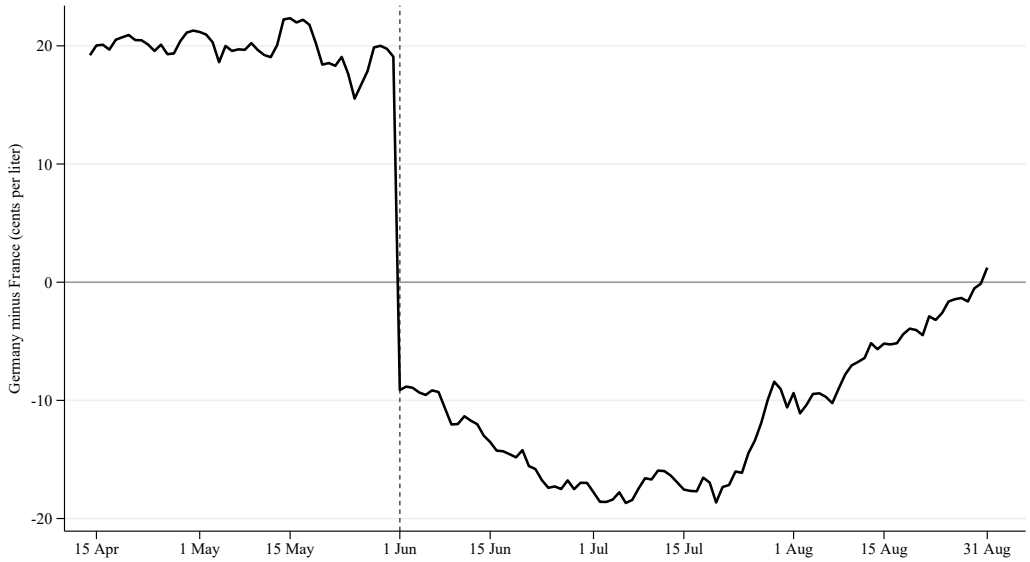


Figure B.2. E5-price difference between Germany and France

Notes: Daily average price of the German sample stations minus that of the French sample stations, in cents per liter. The vertical line marks 1 June 2022.

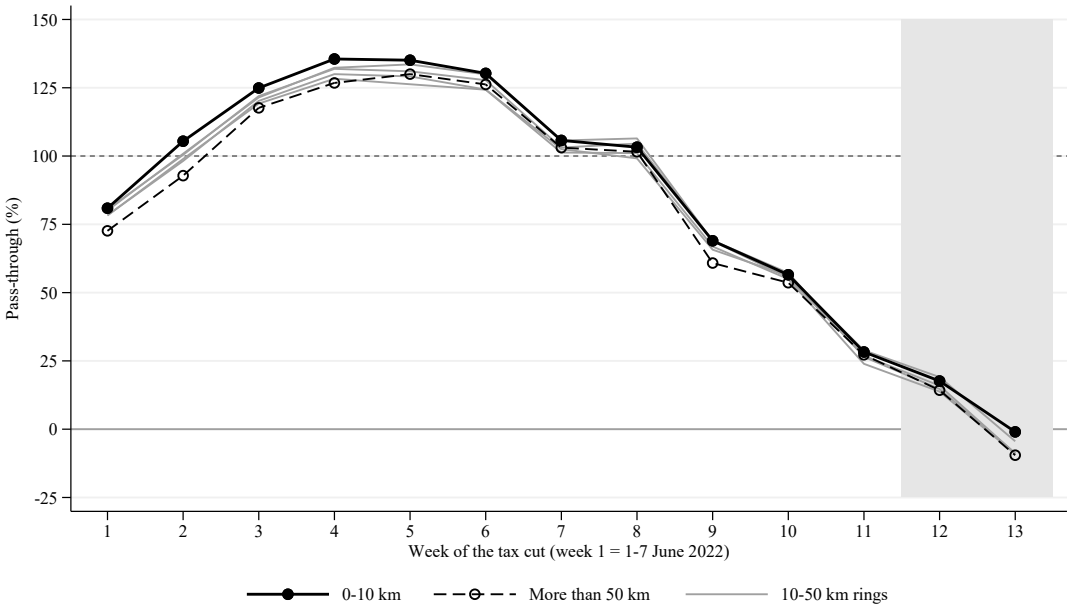


Figure B.3. Weekly diesel pass-through by German distance ring

Notes: Each series is estimated separately for every policy week from the pre-policy period and that week. It is the price change of one German distance ring relative to all French stations, with station and date fixed effects and holiday controls, divided by the 16.71-cent tax-inclusive diesel cut. Week 13 runs from 24 to 31 August. The shaded weeks (17–31 August) precede the expiry of the German cut and the increase in the French rebate on 1 September and are not interpreted.

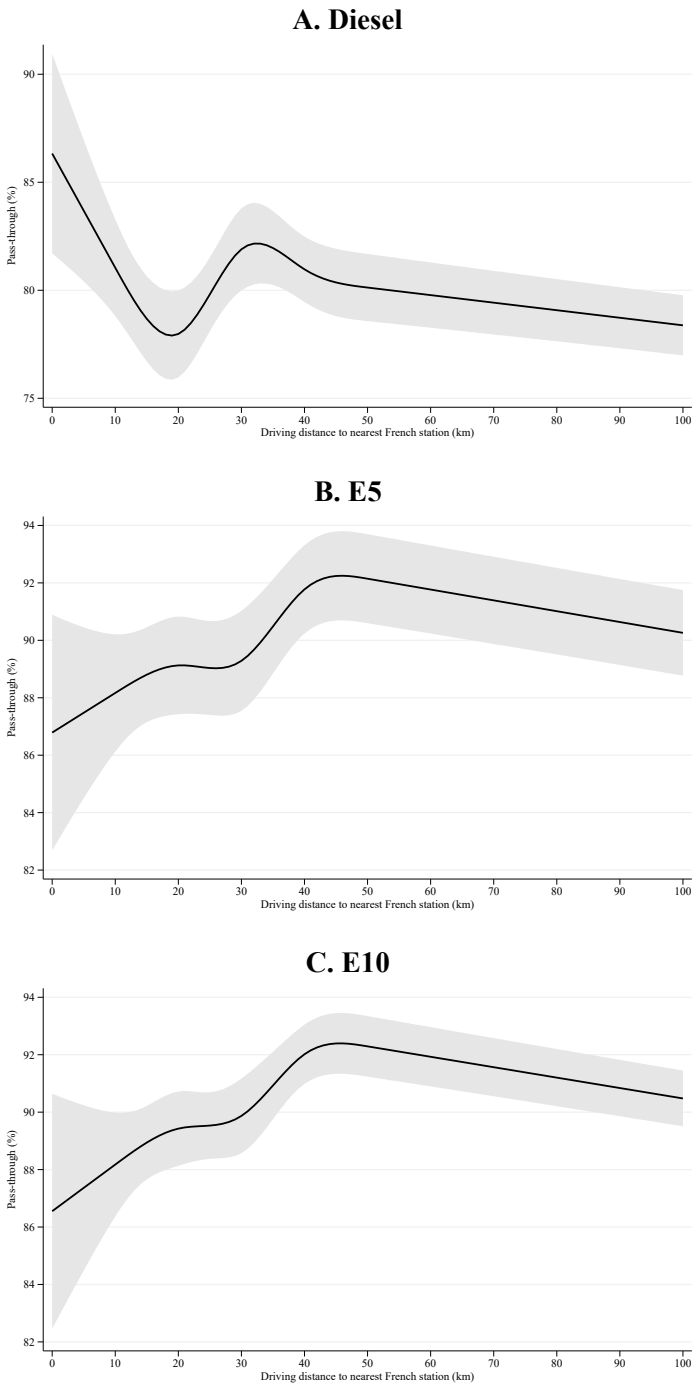


Figure B.4. Predicted German pass-through versus distance by fuel

Notes: Restricted cubic splines use knots at 10, 20, 30, 40, and 50 km. Diesel exhibits a distinct near-border peak; the gasoline curves are nonmonotone and generally rise away from the border. The plotted curves, rather than individual basis coefficients, are the economically interpretable objects.

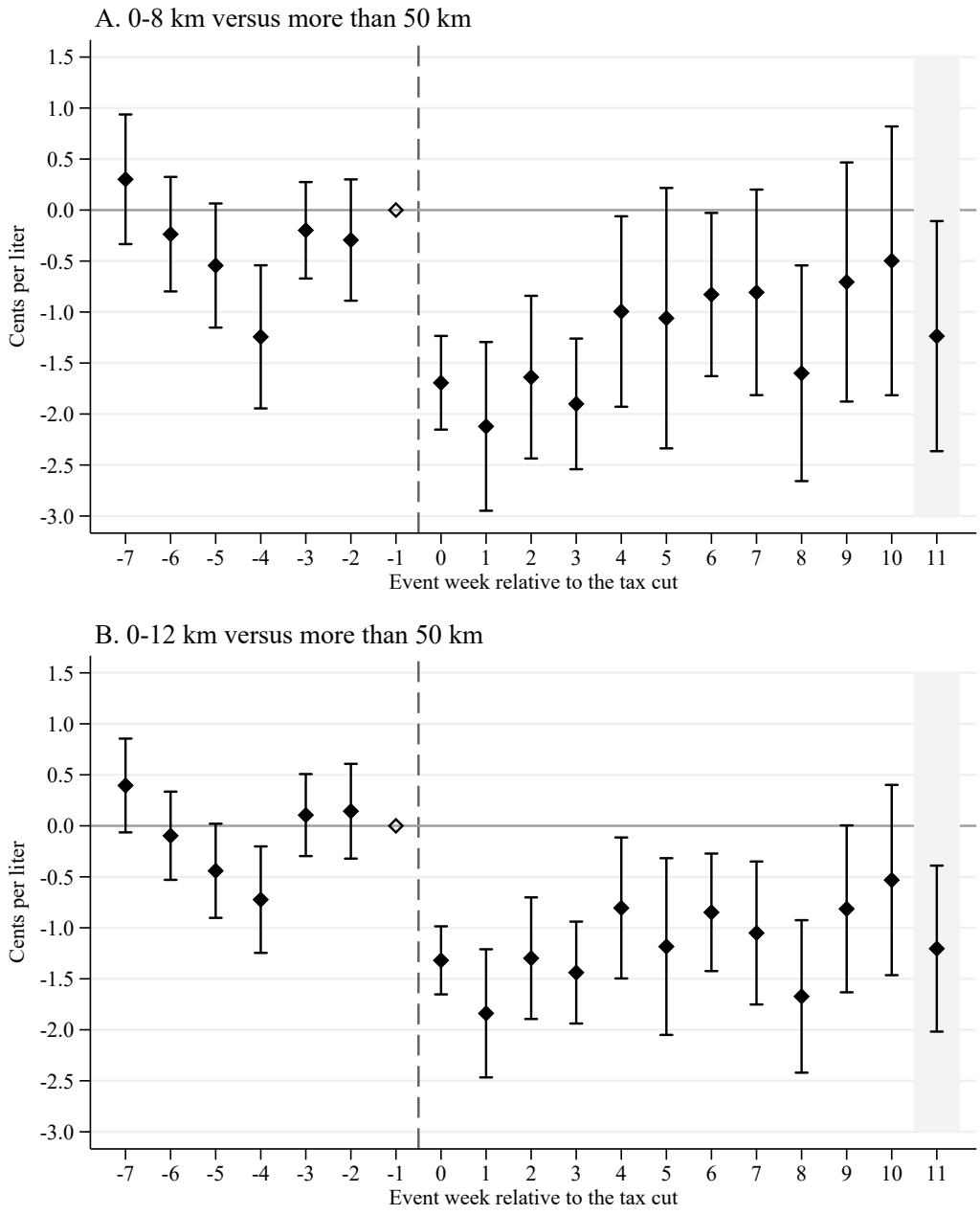


Figure B.5. Event-study sensitivity to the near-border cutoff: German diesel

Notes: Coefficients in cents per liter. Panels replace the primary 0–10 km group with 0–8 km and 0–12 km groups; the comparison group remains German stations more than 50 km from France. Event week −1 (hollow marker) is omitted; the shaded week 11 pools 17–31 August. The near groups contain 53, 67, and 90 stations at the 8, 10, and 12 km cutoffs, respectively. All three specifications reject the joint null for the six pre-policy coefficients and for the post-policy coefficients.

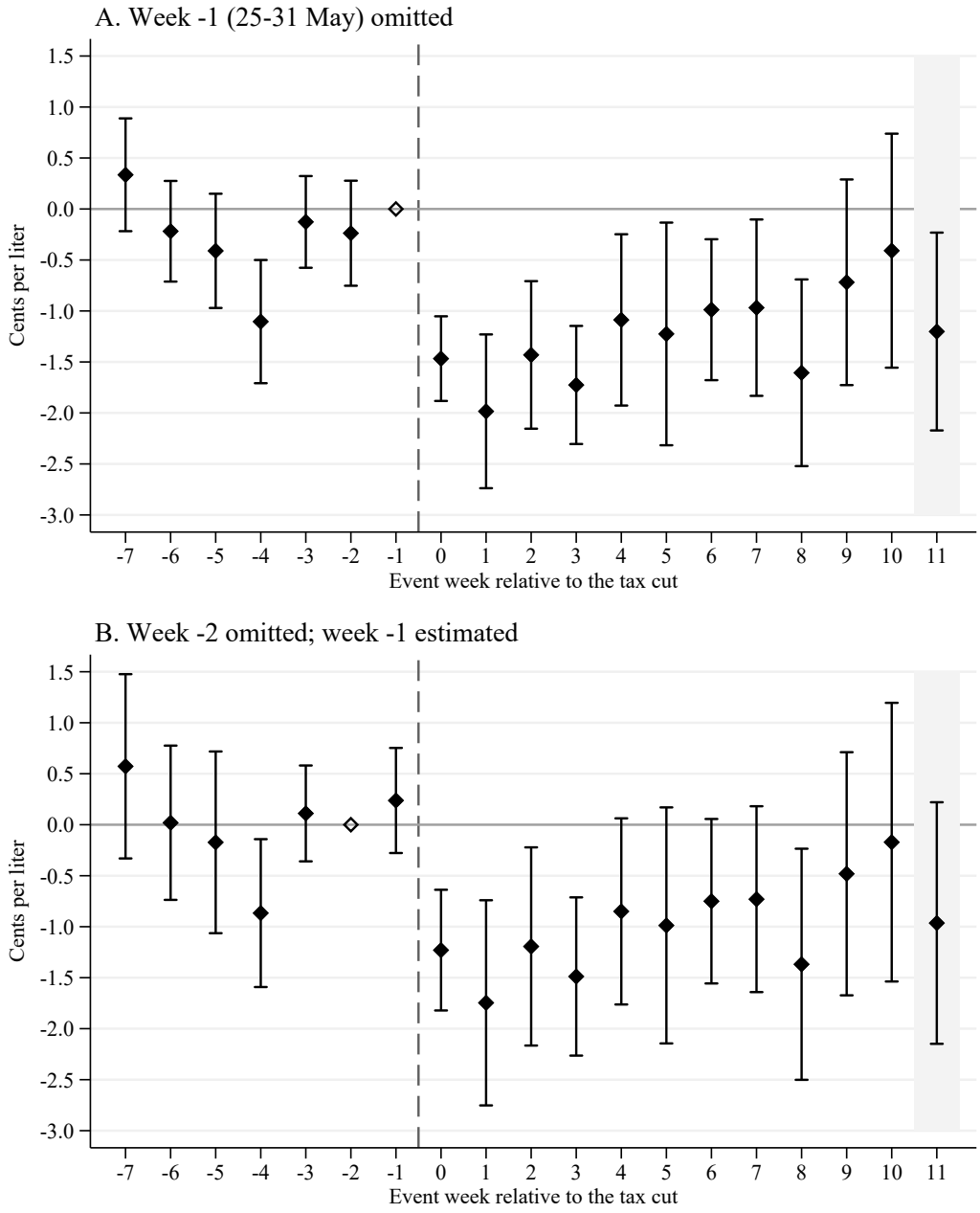


Figure B.6. Event-study normalization and anticipation sensitivity: German diesel

Notes: Coefficients in cents per liter. Panel A omits event week −1 (25–31 May), as in Figure 3. Panel B omits week −2 and estimates the final pre-policy week explicitly; its week −1 coefficient is 0.24 cents (SE 0.26; $p = 0.366$), providing no evidence of a differential last-week anticipation effect. Hollow markers show the omitted week. Re-normalization changes the reference point, not fitted values.

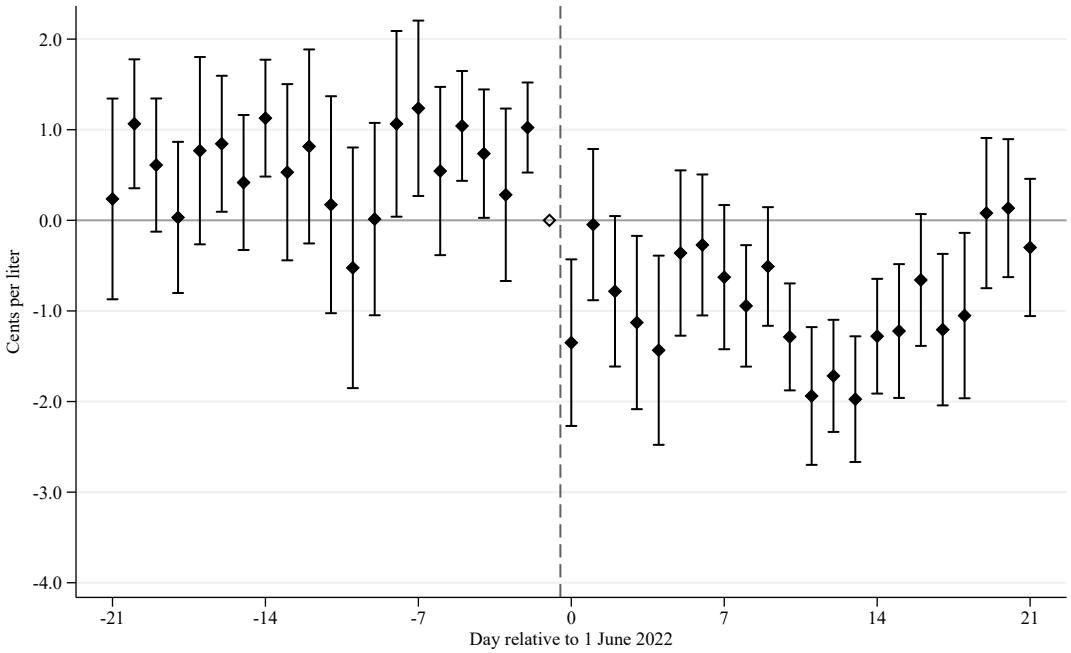


Figure B.7. Daily event study: German 0–10 km diesel ring relative to inland Germany

Notes: Coefficients in cents per liter for days -21 to +21 around 1 June 2022 (day 0); 31 May (day -1, hollow marker) is omitted. Station and date fixed effects and holiday controls; bars are 95% confidence intervals based on station-clustered standard errors. N = 98,218 station-days, 2,285 stations. Relative to the mean of days -21 to -2, the day-0 coefficient is -1.95 cents (SE 0.37).

C Additional tables

Sample and price construction. The French sample covers the ten départements closest to Germany: Moselle, Bas-Rhin, Haut-Rhin, Meurthe-et-Moselle, Vosges, Territoire de Belfort, Meuse, Haute-Marne, Haute-Saône, and Doubs. Driving distances and times use the car profile of the Open Source Routing Machine on a merged OpenStreetMap network for France, Germany, Luxembourg, and Switzerland as of November 2025. The last valid quote observed by 06:00 seeds each station's daily price path, and later quotes update it. The primary sample excludes 201 incomplete station–fuel–days across the underlying panels, including 60 German diesel days. Missing same-day quotes are carried forward from the last valid quote without an age cap; 3-, 7-, and 14-day caps are sensitivity checks (Table C.9). Quotes outside €0.50–€3.00 per liter are removed. Equal timestamps with different prices are treated as ambiguous rather than resolved by file order. Germany has none and France has three; they are excluded from the primary measure, and first- and last-record rules, used as sensitivity checks, leave diesel and E5 unchanged and affect three E10 station-days. A further 1,078 German diesel station-days have no valid routing distance; they enter the descriptive series but not the regressions. All 2,880 German stations are assigned to an urban–rural type; one station 96.7 meters outside a polygon boundary is assigned to the nearest polygon edge. Table C.10 reports the construction of the diesel panel.

Table C.1. E5 descriptive statistics

Country	Period	Mean price	Median	SD	Holiday share	N
France	Pre	1.874	1.869	0.083	0.340	13,757
France	Post	1.973	1.988	0.146	0.613	26,816
Germany	Pre	2.073	2.062	0.082	0.201	135,455
Germany	Post	1.859	1.844	0.106	0.497	261,296

Notes: Prices are euros per liter; N is station-days. Periods as in Table 1.

Table C.2. E10 descriptive statistics

Country	Period	Mean price	Median	SD	Holiday share	N
France	Pre	1.827	1.809	0.093	0.332	28,103
France	Post	1.929	1.937	0.136	0.610	54,399
Germany	Pre	2.017	2.006	0.087	0.200	131,402
Germany	Post	1.803	1.787	0.112	0.497	253,548

Notes: Prices are euros per liter; N is station-days. Periods as in Table 1.

Table C.3. German stations by state and Eurostat urban–rural typology

Panel A. Federal states	
Baden-Württemberg	1,940
Rheinland-Pfalz	773
Saarland	167
Panel B. Eurostat NUTS-3 urban–rural typology	
Predominantly urban	1,022
Intermediate	1,263
Predominantly rural	595
Total	2,880

Notes: Stations are assigned to 2022 municipalities with the VG5000 geometry and then, through the BBSR municipal reference table, to Eurostat's NUTS-3 typology. All 2,880 stations are classified. One station 96.7 meters outside a polygon boundary is assigned to the nearest polygon edge; no station has multiple interior matches.

Table C.4. Spatial price responses: E5

Country and distance	Price change (cents/liter)	SE	*p*	Tax-equivalent response (%)
Germany, >50 km	-32.18	0.29	<0.001	92
Germany, 0–10 km	-31.55	0.56	<0.001	90
Germany, 10–20 km	-31.95	0.32	<0.001	91
Germany, 20–30 km	-32.23	0.33	<0.001	92
Germany, 30–40 km	-32.43	0.35	<0.001	92
Germany, 40–50 km	-31.89	0.37	<0.001	91
France, 0–10 km	-2.95	1.52	0.052	8
France, 10–20 km	-2.28	0.73	0.002	6
France, 20–30 km	-1.91	1.12	0.088	5
France, 30–40 km	-2.76	1.14	0.015	8
France, 40–50 km	-1.87	1.71	0.274	5

Notes: Specification as in Table 2. The final column divides the negative coefficient by the 35.16-cent tax-inclusive gasoline cut; for French rings it is a tax-equivalent scale, not French tax pass-through. N = 437,324. Equality of Germany 0–10 km and Germany more than 50 km: $p = 0.197$.

Table C.5. Spatial price responses: E10

Country and distance	Price change (cents/liter)	SE	*p*	Tax-equivalent response (%)
Germany, >50 km	-31.94	0.22	<0.001	91
Germany, 0–10 km	-31.26	0.54	<0.001	89
Germany, 10–20 km	-31.63	0.29	<0.001	90
Germany, 20–30 km	-32.08	0.26	<0.001	91
Germany, 30–40 km	-32.23	0.29	<0.001	92
Germany, 40–50 km	-31.69	0.32	<0.001	90
France, 0–10 km	-1.16	0.56	0.039	3
France, 10–20 km	-2.19	0.53	<0.001	6
France, 20–30 km	-1.00	0.40	0.013	3
France, 30–40 km	-0.82	0.67	0.226	2
France, 40–50 km	0.51	1.00	0.611	-1

Notes: Specification as in Table 2. The final column divides the negative coefficient by the 35.16-cent tax-inclusive gasoline cut. N = 467,452. Equality of Germany 0–10 km and Germany more than 50 km: $p = 0.176$.

Table C.6. Robustness of the German diesel premium

Specification	Cents	Station SE	County SE	Near stations
Panel A. Specification				
Baseline (German stations only)	-0.97	0.27	0.54	67
Moderators × policy period	-0.76	0.24	0.60	65
County-by-date fixed effects	-0.79	0.38	0.50	67
Government-district-by-date fixed effects	-1.31	0.31	0.60	67
Without motorway stations	-0.95	0.28	0.58	65
Without stations near other borders	-0.83	0.28	0.56	60
Policy period ending 16 August	-0.97	0.27	0.51	67
Comparison group 10–50 km	-0.54	0.28	0.52	67
Panel B. Near-border cutoff (driving distance)				
4 km	-1.70	0.51	0.69	19
6 km	-0.98	0.35	0.42	34
8 km	-0.93	0.30	0.54	53
10 km	-0.97	0.27	0.54	67
12 km	-1.07	0.24	0.52	90
15 km	-0.67	0.19	0.51	137
20 km	-0.37	0.14	0.47	238
Panel C. Triple difference, diesel minus gasoline				
Diesel minus E5	-1.58	0.51	1.34	67
Diesel minus E10	-1.65	0.52	1.38	67

Notes: Each row reports the post-policy price change of German stations within 10 km (Panel B: the stated cutoff) of a French station minus that of German stations more than 50 km away, in cents per liter. All rows include station and date fixed effects and holiday controls. The moderators row interacts the policy period with the pre-reform price, the five largest brands, a motorway indicator, the Eurostat urban–rural type, and the log number of German stations within 5 km; two near stations without a pre-reform price drop out. Government districts are *Regierungsbezirke*. The motorway row is the Table 2 contrast after dropping 37 German motorway stations. The row without other borders drops German stations within 20 km of Luxembourg or Switzerland (driving distance) or Belgium (straight-line). Panel C uses station-by-fuel, station-by-date, and fuel-by-date fixed effects. County standard errors cluster by the 86 NUTS-3 counties.

Table C.7. Placebo treatment beginning 18 May: diesel

Country and distance	Price change (cents/liter)	SE	*p*
Germany, >50 km	0.10	0.17	0.563
Germany, 0–10 km	0.61	0.31	0.050
Germany, 10–20 km	0.18	0.20	0.374
Germany, 20–30 km	0.21	0.19	0.257
Germany, 30–40 km	0.02	0.23	0.925
Germany, 40–50 km	0.12	0.23	0.616
France, 0–10 km	-1.14	0.55	0.039
France, 10–20 km	-1.62	0.54	0.003
France, 20–30 km	-1.03	0.47	0.027
France, 30–40 km	-1.91	0.63	0.002
France, 40–50 km	0.00	0.86	0.998

Notes: Sample: 1–31 May 2022; the placebo period is 18–31 May. Coefficients are relative to French stations more than 50 km from Germany. Station and date fixed effects, holiday controls, and station-clustered standard errors.

Table C.8. Inference for the German diesel premium

Method	SE	*p*	95% interval
Station clusters (2,880)	0.27	<0.001	[-1.49, -0.44]
County clusters (86)	0.54	0.078	[-2.04, 0.11]
County wild cluster bootstrap (86)		0.088	[-2.75, 1.30]
Spatial HAC, 10 km	0.50	0.052	[-1.94, 0.01]
Spatial HAC, 25 km	0.61	0.113	[-2.16, 0.23]
Spatial HAC, 50 km	0.66	0.142	[-2.26, 0.32]
Spatial HAC, 100 km	0.72	0.177	[-2.37, 0.44]

Notes: The estimate is the German-only premium of -0.97 cents per liter (German stations within 10 km of France minus those more than 50 km away; station and date fixed effects, holiday controls; N = 400,250). The wild cluster bootstrap uses Webb weights and 9,999 draws on a regression with date indicators and a station effect. Spatial HAC standard errors use a Bartlett kernel over station distance with the stated cutoff and full serial correlation within stations.

Table C.9. Price-construction sensitivity: tax-equivalent diesel response (%)

Group	Primary	00:00–24:00	3-day cap	7-day cap	14-day cap
Germany, >50 km	79.1	78.0	83.1	80.9	80.5
Germany, 0–10 km	84.9	83.5	89.6	87.2	86.6
France, 0–10 km	4.8	4.9	7.3	5.9	5.2
France, 30–40 km	5.2	5.3	3.5	4.2	4.3

Notes: The primary price averages 06:00–22:00 and has no cap on the age of a carried-forward price; the cap columns drop station-days whose price is older than the stated number of days. Sample sizes are 507,791 (primary), 507,790 (00:00–24:00), 450,130 (3 days), 485,267 (7 days), and 498,864 (14 days).

Table C.10. Diesel panel construction

Construction step	Station-days
German complete 06:00–22:00 station-days	401,328
of which using a carried-forward price	29,502
German days excluded for missing route	1,078
German estimation sample	400,250
German partial-coverage days excluded	60
French estimation sample	107,541

Notes: The German complete-day count precedes routing exclusions. Across Germany and France, the raw files contain 25,484,414 valid price events. France has three ambiguous timestamps (six rows); Germany has none.